



Show that the eigenvalues of a unitary matrix have modulus 1

Asked 8 years, 8 months ago Modified 8 months ago Viewed 78k times



Show that the eigenvalues of a unitary matrix have modulus 1.

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I know that a unitary matrix can be defined as a square complex matrix A , such that

$$AA^* = A^*A = I$$



where A^* is the [conjugate transpose](#) of A , and I is the identity matrix. Furthermore, for a square matrix A , the eigenvalue equation is expressed by

$$Av = \lambda v$$

If I use the relationship $(uv)^* = v^*u^*$ and take the conjugate transpose of this equation then

$$v^*A^* = \lambda^*v^*$$

But now I got stuck. Can someone help?

linear-algebra

matrices

proof-writing

eigenvalues-eigenvectors

unitary-matrices

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edited Apr 7 at 8:10



Sandeep Tiwari

646 3 23

asked Mar 28, 2016 at 20:39



Alim Teacher

1,273 5 22 30

i think you can find your answer in part 4 of this post math.stackexchange.com/questions/40246/...
– K.K.McDonald Mar 28, 2016 at 20:46

I actually looked at that post but it was difficult to understand. I think they do it in another way.
– Alim Teacher Mar 28, 2016 at 20:51

4 Answers

Sorted by: Highest score (default)



You multiply your two relations to obtain

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$$\begin{aligned} v^* A^* A v &= \lambda^* v^* \lambda v \\ v^* I v &= (\lambda^* \lambda) v^* v \\ v^* v &= (\lambda^* \lambda) v^* v \\ \|v\|^2 &= |\lambda|^2 \|v\|^2 \\ \sqrt{1} &= |\lambda| \\ 1 &= |\lambda| \end{aligned}$$

Recall that the [modulus of a complex number](#) $\lambda = a + bi$, also called the "complex norm", is denoted $|\lambda|$ and defined by $|\lambda| = |a + bi| = \sqrt{a^2 + b^2}$ and $\lambda^* \lambda = (a - bi)(a + bi) = a^2 + b^2$. Hence $\lambda^* \lambda = |\lambda|^2$.

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edited Mar 11, 2018 at 13:16
user168764

answered Mar 28, 2016 at 20:52



Bérénice

9,457

2

26

37

Why is the negative square root in the end neglected ? Shouldn't it be $\lambda = +1, -1$ – Shashaank Mar 3, 2021 at 12:58

2 Sorry for the late answer, it is because a modulus is always positive – Bérénice May 23, 2021 at 6:38

1 Very good proof! However, an interesting thing is that you can perhaps stop at the third last step, because an equivalent condition of a unitary matrix is that its eigenvector lies on the unit circle, so therefore, has magnitude 1. – Danish A. Alvi Mar 8, 2022 at 17:58



12

A unitary matrix U preserves the inner product: $\langle Ux, Ux \rangle = \langle x, U^* U x \rangle = \langle x, x \rangle$.

Thus if λ is an eigenvalue, $Ux = \lambda x$, we get $|\lambda|^2 \langle x, x \rangle = \langle \lambda x, \lambda x \rangle = \langle Ux, Ux \rangle = \langle x, x \rangle$.

So $|\lambda|^2 = 1 \implies |\lambda| = 1$.



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answered Jul 2, 2019 at 19:12

user403337

Why is the negative square root in the end neglected ? Shouldn't it be $\lambda = +1, -1$ – Shashaank Mar 3, 2021 at 12:57

@Shashaank The modulus is always nonnegative. – user403337 Mar 3, 2021 at 18:09

$|\lambda| = 1$ implies that λ lies on the unit circle. @Shashaank – user403337 Mar 3, 2021 at 18:35 ✎



4

Δ as λ

$Av = \Delta v$



$(Av)^* = (\Delta v)^*$

$v^* A^* = \Delta^* v^*$

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$$v^* A^* A v = \Delta^* v^* \Delta v$$

$$\text{As } A^* A = I$$

$$v^* I v = \Delta^* \Delta v^* v$$

$$v^* v = \Delta^* \Delta v^* v$$

$$(1 - \Delta^* \Delta) v^* v = 0$$

Since v is not equal to zero.

Hence

$$1 - \Delta \Delta^* = 0 \implies \Delta^* \Delta = 1$$

$$|\Delta|^2 = 1 \implies |\Delta| = 1.$$

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edited Aug 15, 2018 at 17:37



Daniel Buck

3,638 1 14 28

answered Aug 15, 2018 at 17:15



Stark Clark

41 2

What was the point changing λ to Δ ? See meta.math.stackexchange.com/questions/5020/... for help formatting in MathJax. – Daniel Buck Aug 15, 2018 at 17:31

I've added formatting to your answer, hope that's ok. Note that two * format as italic in HTML which was why some stuff rendered funny.. – Daniel Buck Aug 15, 2018 at 17:39



The result that you seek follows from the following.

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Lemma. If A is unitary and $\|x\|_2 = 1$, then $\|Ax\|_2 = 1$.



Proof. By definition,

$$\|Ax\|_2^2 = \langle Ax, Ax \rangle = (Ax)^*(Ax) = x^* A^* A x = x^* x = \|x\|_2^2 = 1.$$



If (λ, v) is an eigenpair of A , then, without loss of generality, it may be assumed that $\|v\|_2 = 1$. Following the lemma above and the absolute homogeneity property of vector norms, notice that

$$|\lambda| = |\lambda| \|v\|_2 = \|\lambda v\|_2 = \|Av\|_2 = 1,$$

as desired.

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edited Jul 2, 2019 at 18:07

answered Mar 20, 2019 at 22:04



Pietro Paparella

3,580 1 20 30